

Chapter 6

University-Industry Alliances and Consortia



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6.1 Introduction

Research-focused alliances between companies and universities are integrated partnerships that leverage multiple types of transactional and strategic collaborations in support of shared goals. Oftentimes, University-Industry (UI) alliances start as transactional or one-off relationships, where a company sponsors research in a single university lab. Then, over time, the relationship grows to be more involved. Instead of focusing on a single lab on campus, the company may invest in research across multiple labs or departments.

Another method for collaboration allows the company to proactively work with the university to recruit or train students through undergraduate capstone projects, MS or PhD level internships, or supporting extracurricular experiential learning programs. In addition to sponsored research and student engagement, the company and university may exchange access to specialized materials or facilities or work together to pursue government funded opportunities. Taken together, these activities support a strong, integrated UI alliance.

When a UI relationship grows to be larger than a single researcher or lab, and it encompasses multiple parts of the university and company, then the relationship is often considered an alliance and is managed more centrally in addition to the management of individual projects or initiatives. While a UI alliance does not need to follow this trajectory, it is common for more limited interactions to be the precursor

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to more involved, multifaceted partnerships. Such an approach allows for trust to be built and shared goals to be identified.

Consortia involves one or more universities and one or more companies committing resources (time, financial, materials, and facilities) to support a shared interest. Consortia can be university-led or company-driven. This chapter will focus primarily on university-led consortia with research agendas that are driven by industry needs and active engagement, because they are the most pervasive style of consortium among UI research or innovation driven partnerships.

While countless types of UI engagements can support multifaceted research- and innovation-driven partnerships, this chapter focuses on a select set of collaborative models that progress from somewhat hands-off to more integrated and involved partnerships. The examples selected for this chapter build on one another to demonstrate how different collaborative models can be put together to support truly strategic alliances and integrated consortia. The discussion is motivated by the idea that UI alliances and consortia can be a strong economic driver by connecting research to innovation to production and job creation. Later in this chapter, we explore one successful partnership that emerged into a robust consortium with other corporations adding to the research.

This is the first chapter in the book section on Avenues for UI engagement. Within this and the following seven chapters, the book addresses the key ways in which UI engagement occurs, addressing strategic alliances and consortia, innovation centers and economic development, entrepreneurship, executive education, academic health systems, and clinical trials. The section concludes with strategies for gaining support for UI engagement, first within companies and second within universities. As shown in Fig. 6.1, the Avenues section builds from the foundations presented in the first four chapters, and provides a precursor to the final section on the mechanics of UI engagement.

		Mechanics
Foundations	Avenues	
Alignment and Strategy	Alumni and External Engagement	Commercialization
Barriers and Opportunities	Clinical and Sponsored Research	Copyrights
Collaboration	Community Partnerships	Data Security and Sharing
Engagement	Co-operative Education	External Engagement
Grand Challenges	Corporate Venture	Intellectual Property
Innovation and Interfaces	Economic and Regional Development	International Collaborations
Organizational Culture	Entrepreneurship	Licensing and Patents
Partnerships	Funding and Support Initiatives	Material Transfer
Roadmaps	Industry Partnerships	Policies and Regulations
Stakeholders	Research parks	Privacy and Transparency
	Startups	Publicity
	Talent and Skill Development	Research Ethics
		Responsible Conduct of Research
		Training and Professional Development

Fig. 6.1 Key avenues for UI engagement

6.2 Opportunities and Challenges

UI partnerships, via alliances and consortia, provide many benefits—including resources, guidance, and funding—that help universities innovate for next generation technologies. Unfortunately, alliances and consortia have not yet met their full potential. Industry funding constitutes only a small percentage of total university research funding—a mere 6% according to a 2024 report from the Association of Public and Land Grant Universities (APLU). With greater attention, alliances and consortia offer an avenue to grow funding in support of university and industry objectives. Universities can be better aligned to industry needs. Industry can gain access to university research innovations and talent. In the process, alliances and consortia can provide real-time closed loop feedback for enhancing research, furthering innovation in companies and universities alike.

In the simplest terms, partnerships allow universities to become closer to industry priorities, research assets, co-investment, and alumni networks. Often focused on a specific technical area or technology, a professor or department becomes engaged with a corporation to do specific work, either for applied technologies to meet technical needs or problems or more theoretical research to support emerging markets or technology needs. Often these relationships follow an employee's or executive's previous experience, either through study or a conference, and continue with additional research directed to the area of interest from the corporation.

Perhaps the most difficult part for universities is understanding the needs of varied companies, and the needs of the varied people working within companies. A company representative may be looking for:

- Research (basic & applied)
- Engineering Expertise—Technology Problem Solving, Testing, Use of Cutting-edge labs
- IT Support—Infrastructure, Security, AI
- Business Development—Relationships
- Manufacturing—Industrial Solutions, Robotics
- Human Resources—Talent Pipeline Workforce, Industrial Relations
- Business—Market Access, Customers
- Marketing—Relationship Marketing, Reputation

The corporate ecosystem is continually evolving with organizational realignments, new technologies, changing customer needs, products, and changes in business environments. Maintaining long-standing partnerships is hard work. This challenge is exacerbated by complex industry organizations, including senior leadership, engineering personnel, finance cycles, legal requirements, competition, and community engagement.

Perhaps the most significant challenge is the continuity and continuation of collaborative interactions between a company and university when the individuals who are key to the relationship switch to different jobs or different job descriptions. Even changes in contracting or executive branch personnel can quickly disengage a

formal long-term relationship. It is critical that the university has a succession plan for personnel, and that they understand emerging business pressures. Often, organizational changes at a company occur quickly and are not understood by the university. Changes in personnel or in organizational alignment can interrupt funding cycles, reporting structure, and priorities. Often, university research priorities must be realigned and changed according to a business cycle or funding challenges. Key elements for success are value propositions, relationship managers and legal agreements, as described below.

Value Proposition UI alliances and consortia should showcase their current engagement and successes. Therefore, universities must provide a clear executive level value proposition with associated deliverables and metrics. Sharing research details and conveying the value proposition are equally important. The proposition needs to define value in terms of the corporation or community, not just the university. Often the industry technical sponsor and the university researcher understand the research and the associated value but the true industry ownership of the budget is at a business unit or higher executive level. The senior executive may make a quick decision based upon their understanding and assessment of the value. Universities should imagine that senior leadership of the company needs to digest the value proposition in the time it takes to complete a short elevator ride.

University research is generally one of two types, basic or applied. Basic research is identified as having a low technology readiness level, meaning that it will take additional development before being ready for application. This research is often completed by professors working on theoretical research. Often industry partners have groups of engineers or scientists that are exploring this research for future products. Sometimes this research is completed by a centralized R&D organization. Applied research is generally more closely related to the product. It can be for direct use by a product line or an industry supplier. Understanding the interest of students or faculty is critical when a company is presenting the value proposition for the university to work with them. The value proposition for a company to engage in an alliance or consortium with one or more universities is often different from the value proposition for the university's participation. But, successful UI partnerships are mutually beneficial and the value proposition for participation is a key component of demonstrating and communicating the benefits of the collaborative activities.

Relationship Managers For the purpose of UI partnerships, relationship managers are the key individuals at both the company and university who take ownership of coordinating collaboration between the partners. Relationship managers may be individuals in a centralized, formal role, or the main point of contact may be a researcher or someone embedded in a company business unit or a university department or center. Regardless of where they sit in the company or the university, dedicated relationship managers need clear responsibility. Establishing an alliance with another organization without a clear plan and long-standing relationships is almost impossible. Relationships can come from steering committees, individual transactions, such as material transfer agreements or facility-use agreements, or individual

sponsored research projects. Relationships must be maintained through regular, nonnegotiable meetings. Dialogues must ensure the sustainability of these relationships. A portion of relationships management is also understanding the industry calendar for budgeting and decision making, which can conflict with the university calendars. The partnership modalities discussed later provide examples of the partnership infrastructure that can be used to support sustained dialogue among UI partners.

Both universities and industry have multiple levels of organization that are responsible for engagements. Until recently, university roles and responsibilities have been more stable than industry partners. However, today's environment of distribution and discontinuity requires vigilance, both for the industry researcher and the university, to understand the changing environment and funding priorities. Companies often involve individuals from contracting, legal, marketing, and engineering in decision making. The relationship manager must understand the ecosystem and be aware of funding cycles and changes.

Legal Agreements Another major success enabler is the establishment of master agreements. Industry can choose to walk away from the engagement if the Intellectual Property (IP) and other important terms, such as liability, defined in the agreement are not quickly implemented, especially when funding cycles for corporations are well defined. Additionally, excess unspent money at the end of a year or a budget underrun can be contracted to universities if a master agreement has already been executed. These agreements must document what the host and members are committing to when they join the collaboration. Ideally, the master agreement remains unchanged over time, with the exception of a review every few years to ensure that it aligns with any new laws or changes in partnerships may have undergone. The master agreement should cover:

- Research objective and project types
- Membership types
- Management structure
- Use of contributions
- Meeting types and schedules
- Duration and renewal
- Termination
- Publication
- IP ownership and use
- Confidentiality
- Liability
- Export control
- Publicity
- Governing law
- Notices
- Additional terms deemed appropriate by the host and members

6.3 Prior Research

The principles of strategic planning and partnership building apply to UI alliances and consortia. Countless resources have been developed over the years on these topics. For example, the McKinsey 7S framework for strategic planning focuses on strategy, structure, systems, shared values, skills, style, and staff (Peters & Waterman, 1984). Although the 7S framework was conceived as a strategic planning tool for companies to use internally, the principles hold true for UI alliances and consortia. The case study at the end of the chapter shows the 7S's in action. Similarly, groups like Partnernomics emphasize executive-level buy-in, setting clear goals, and dedicating appropriate resources (Partnernomics, 2022). These principles map back to the key elements for success discussed in the introduction.

6.4 Options, Solutions, and Impact

This section provides four approaches for forming alliances and consortia: (1) Industry advisory boards, (2) Corporate Affiliate Programs (CAPs), (3) University-based research centers, and (4) Innovation hubs. Each example is described from the perspective of value proposition, relationship managers, and legal agreements.

6.4.1 *Industry Advisory Boards*

At their core, UI partnerships are not defined by large impersonal organizations. They are built on the trust and relationships of individual people. Efforts, such as advisory boards, can be used to institutionalize organization level relationships and ensure that connections are established and maintained. Industry Advisory Boards (IABs) are one of many mechanisms to foster personal relationships between the company representatives, academic researchers, and research leadership who seek to foster impact-driven collaborative partnerships.

IABs are a relatively low effort, high return, method for engaging industry representatives in university-based research projects. An IAB is a collection of industry representatives who meet on a recurring basis to provide input into the research directions and talent development priorities of the university, college, department, or center. IABs create pathways for communication that provide the necessary alignment for a strategic partnership to be fruitful.

While an advisory board is not a consortium that produces research or workforce development programs, it is an opportunity for serendipitous interactions that can lead to more impactful, outcome-oriented collaborations. Typically, the companies represented on an advisory board are interested in maintaining or growing their involvement in the overall university's success, sponsored research, experiential

learning, recruiting, curriculum development, or other activities on campus. For the companies participating in the advisory board, IABs are an opportunity to convey their company's needs and to have "insider" knowledge of the university's activities. For a university, advisory boards can provide valuable input that allows the university to make informed decisions about how the college, department, or research center's staffing, programming, and research align with an industry perspective. This can result in proposing research directions that industry is ready to fund, and in the creation of student programs that result in their students being easier to place into careers with companies on the IAB. To be successful, an advisory board should meet on a recurring basis and their recommendations should be acted upon. Even if a university department or research center chooses not to follow the recommendations of the IAB, communicating that decision back to the board creates good will and maintains the buy-in of the board.

6.4.2 Corporate Affiliate Programs (CAPs)

Corporate Affiliate Programs (CAPs) can take on a variety of forms. For the purpose of the discussion here, the focus here will be on university-led programs where multiple companies participate in a forum focused on fostering research in a specific area, connecting with students, or more deeply connecting with the academic community. The CAP allows company representatives, students, faculty, and university personnel to identify real-world problems and mobilize resources to tackle those challenges. When a CAP program has a clearly defined value-proposition that simultaneously speaks to the company's internal priorities and the university's research strengths, then a CAP can be valuable for identifying shared goals and advancing research, innovation, and/or workforce development. The key principles highlighted in the opportunities section are applicable to CAPs as well, as described below.

Value Proposition CAPs are most successful when they have a clear value proposition that speaks to both the strengths of the university and the needs of potential corporate partners. To succeed, a CAP must be more than a fundraising venture. It must instead be motivated by the value derived from research, learning experiences, workforce development, and innovation. Examples of research-focused metrics used to evaluate CAPs include the number of dollars spent or leveraged from other sources, the amount of sponsored research that is conducted as follow-on projects to research that originally was done as a part of the CAP, or the number of students engaged in research projects through the CAP. When evaluating whether to join a CAP, companies often consider the following:

- Alignment between the company's research priorities and the focus of the CAP
- When the CAP was created and how well established is the CAP
- Past successes of the CAP, such as invention disclosures or patents

- Opportunities to connect with others in the innovation ecosystem, such as companies interested in similar research for different applications from the applications their own company is considering
- Depth of engagement with individual faculty members or students

Meanwhile, the university may consider the number of corporate representatives who are active in the CAP and who become connected to campus in other ways as a result of their engagement. From the perspective of student support, metrics such as the number of internships or number of full-time hires, may be considered.

Relationship Managers For a CAP to be successful, dedicated personnel need to be identified and committed from all participating organizations. For the university, this means someone is assigned to manage the CAP—scheduling meetings, following up with corporate partners, organizing faculty to present, and following through to mobilize research or student programming. From a company perspective, this means assigning a person to own the relationship with the CAP. The company representatives engaged in CAPs also have a responsibility to connect the faculty, research, or students that they work with through the CAP to the appropriate parts of the company. Oftentimes, multiple parts of a company may benefit from collaborating with a university. Taking advantage of that opportunity requires internal coordination on the part of the company's relationship manager.

Legal Agreements CAPs can be gift based or contract based. If a CAP is gift-based, then the corporate members of the CAP have less direct influence over how the money is spent and may gain less benefit, since the results of research performed from a corporate gift may not be directly commercially exploited. Gift agreements are less involved than contracts, because a direct deliverable, such as a report or IP, are not promised. This can make the agreement process move more quickly, but it can also diminish the value that a company perceives from a research-based CAP. In the case of a research-focused CAP, if IP is expected to result from the funded research, then a contract-based approach is necessary. Oftentimes, CAPs pool funds from corporate partners. The corporate partners identify which research projects to fund with the pooled funds, and then they negotiate a license to the Foreground Intellectual Property (FIP) after the research project concludes. In the event that none of the CAP members pursue a license, the university would retain the right to license the FIP to another party. Additionally, it is common for members of a CAP to fund an initial project in a particular area. Then, if the results are promising, an individual company would sponsor focused research outside the CAP. Gift-based CAPs may make sense if the CAP is more student and talent pipeline focused as opposed to research focused.

6.4.3 *University-Based Research Centers*

University-based research centers can be housed within a department or college. Alternatively, they can be interdisciplinary, spanning multiple colleges or may include faculty or resources at multiple universities. In some cases, a university research center may receive some of its funding from the government, such as the Industry-University Cooperative Research Center and Engineering Research Center programs through the U.S. National Science Foundation or the Manufacturing USA Institutes, which is run through an office headquartered in the U.S. Department of Commerce.

University research centers can have a focus on basic research or can be more applied in nature. But in all cases, they leverage a myriad of resources from the organizations. A university research center that engages a consortium of other universities and companies may engage students and faculty through research, experiential learning, access to specialized facilities, and workforce or training initiatives. These centers may also be recipients of in-kind contributions from their corporate partners, potentially including the time and expertise of industrial researchers, novel materials, software, and hardware.

Value Proposition Usually, the university selects the research focus of a research center. Then, industry members join the consortium based on alignment between the company's internal needs and the research or workforce focus of the center. Typically, a university center also allows industry members to access a larger pool of research than they would have access to by funding a single research project or single lab.

Relationship Managers Oftentimes, university research centers are managed by university staff. These staff may play other roles within the university, such as corporate engagement, or they may be faculty members. From the industry side, participating in the university center may be an assigned part of an employee's job, but it can also be a "volunteer role," meaning they are not rewarded for participation and have other assigned job functions within the company. Regardless of how the center is managed, collaborators should have regular opportunities to exchange ideas, to ensure mutual benefit.

Legal Agreements Different research centers deploy different IP approaches. Many consortia give the industry members the first right of refusal for access to foreground IP. This allows companies that invested in the research the option to license the resulting technology before the university approaches other companies about licensing it. Depending on the consortium, multiple companies may be able to exercise exclusive licenses within a field of use. However, for some consortia, it is predetermined that all foreground IP will be put in the public domain or made open source. Public dedication or open-source IP approaches are more common in fields like software than they are in fields that develop tangible goods, such as prototypes or unique molecules.

6.4.4 *Innovation Hubs*

Cross-sector partnerships play an increasingly vital role in driving innovation and for societal level impact. Innovation is the creation and adoption of new solutions that solve problems and add value. UI alliances and consortia play a critical role in leveraging use-inspired research to drive innovation. Incorporating an industry perspective in academic research increases the likelihood research results in goods and services that better society and have lasting economic impact.

Increasingly, the connection between innovation and economic development has motivated increased investments in research- and innovation-focused ecosystems. These ecosystems are interconnected, dynamic networks of stakeholders and resources focused on catalyzing the discovery, development, and deployment of new technologies and the workforce to support them (UIDP, 2024). Place-based innovation hubs organize key stakeholders from across the research and innovation spectrum and create a common vision that couples their region's unique strengths with a common vision and purpose.

Innovation hubs are one of the most complex and involved versions of a cross-sector consortium. Other types of consortia may focus on specific directions, such as fueling research in an area of industrial relevance, developing workforce and talent pipeline programs to ensure a skilled workforce, or translating research from lab to market. Many innovation hubs seek to bring together all of these objectives, often in conjunction with programs aimed at startups, small and medium-sized enterprises, and manufacturing. Key players in an innovation hub often include:

- Research organizations, such as universities, national labs, and nonprofits
- Companies of all sizes, including suppliers and startups
- Investors ranging from state and national governments to venture capital to corporate or philanthropic organizations
- Commercial real estate developers
- Connectors to organize and manage the hub

Successful innovation hubs are built on a common vision with substantive investment and buy-in from a small, select group at the onset. In some cases, the initial investment may come from a large company. In other cases, an infusion of either state-level or national-level dollars is the catalyst for the creation of the innovation hub. In all cases, successful innovation hubs are built on a foundation of regional strengths around key areas, such as research reputation, specialized facilities, available workforce (even if reskilling or upskilling is necessary), and enthusiasm to work collectively with key players within the region. The key elements to success, highlighted earlier in this chapter, are especially applicable to innovation hubs.

Value Proposition First and foremost, the organizations contributing to an innovation hub need to see the direct return from their investment. Investments in innovation hubs tend to be large-scale, often involving building facilities, hiring dedicated staff, and developing processes for working more closely with a wide range of partner organizations. As a result, the potential pay-off from participating in an

innovation hub needs to be compelling. For a company, this may mean the ability to couple the building of specialized research or production facilities with manufacturing capacity and a talent pipeline to ensure that appropriately trained workers are present in the local area. For a university, the value proposition for an innovation ecosystem may be focused on student success, community engagement, or elevating the university's profile. Local or state governments are often interested in job creation and the amount of investment in the local area. Meanwhile, small and medium-sized companies may be interested in an available customer base or demonstrating value to potential investors.

Relationship Managers Innovation hubs require buy-in and active participation of the core stakeholders. This buy-in is not restricted to the commitment of financial or other resources. People within each participating organization must be identified and they must participate in regular, recurring meetings to ensure continuous alignment on priorities and programming to support those goals. Key personnel need to take ownership of the relationship. Innovation hubs are too complex and too involved to assume that everyone will organize themselves. Rather, each key player in the innovation hub needs to assign one or more people to manage the relationship, coordinate with other key players, and ensure that the goals of the innovation hub are supported by their organization.

Legal Agreements Formal legal agreements ensure that each stakeholder understands what is expected of them as a participant, their financial commitments, and how their IP may or may not be used. It is generally advisable for those who are most invested in the innovation hub to sort out legal and financial expectations, including signing the appropriate agreements, prior to engaging additional partners. It is inefficient and impractical to negotiate separate terms and conditions with every contributor to an innovation hub. In addition to legal agreements, a governance structure needs to be designed to engage the key players in ways that drive the hub toward the stated objectives.

6.5 Case Study

The Advanced Manufacturing Research Centre (AMRC) at the University of Sheffield in Sheffield, UK, is an ideal example of effective alignment of university and industry research objectives along with government economic development goals. The AMRC was first conceptualized in 1999 by its co-founders, Professor Keith Ridgway at the University of Sheffield and Adrian Allen, who at the time worked for a custom machining tool manufacturing company based in Sheffield. Allen and Ridgway approached The Boeing Company in the fall of 1999 with a proposal to perform research in the area of hard metals machining (titanium and super alloy steels) at the University of Sheffield—a research area in which the university had clearly demonstrated international leadership. The proposed approach

reasoned that instead of many aerospace companies independently investing research funds trying to increase the efficiency of machining of hard metals, it would be better to form a group at the University of Sheffield to co-invest in the research, leverage other companies' investments and pursue government grants. This concept caught Boeing's interest, and after a series of meetings and working sessions to finalize a collaboration agreement, the Advanced Manufacturing Research Centre with Boeing was launched in 2001.

At launch, Boeing was AMRC's only industry member. Use of the Boeing trademark in the title of the research collaborative was impactful in the early stages of the consortium, since the brand attracted other major aerospace original equipment manufacturers (OEMs) as well as their supply chains. The proven expertise of the University of Sheffield in hard metals machining combined with a multi-year research commitment by Boeing broke free a significant UK government grant for the AMRC to stand up their first 10,000 square foot building and fill it with state-of-the-art machining equipment, plus hire a small full-time research staff. The initial success of the Boeing engagement allowed the AMRC to rapidly recruit new industry members, each bringing a multi-year commitment of research funding. The AMRC continued to grow and expand over the following two decades. As of 2024, the AMRC employs 600+ world leading researchers, technicians and support staff, engages over 120 industry members from global OEMs to local SMEs and manages a 40 million GBP annual research portfolio. In its lifetime, AMRC has attracted an additional 300 million GBP investment in the community surrounding the research campus.

At its launch the AMRC was part of the faculty (school) of engineering of the University of Sheffield, but with its growth is now an independent faculty of the university. The AMRC location is a 20-minute drive east of the main campus at the site of a former open coal mine producing coke for steel making. This geographic location provides space to erect large buildings filled with industry-scale equipment and creates a commercially oriented sense of place important to attracting industry engagement. In addition to location, hiring full-time research staff has been an important factor in moving at the speed of industry while maintaining ties to the academic community on the main campus. Unlike traditional university research, this arrangement allows professors and graduate students to operate on their academic calendar and schedules while the AMRC is able to offer timely, year-round support to industry members.

The success of the AMRC can be attributed to a well-designed collaboration agreement. The agreement features a two-tier membership model, appropriately sized for large and small companies. Members make a multi-year financial commitment and IP rules are clear and balanced. Tier 1 members were asked to commit to an on-going research investment of 200,000 GBP per year, while Tier 2 members contributed 30,000 GBP per year. In-kind contributions of equipment and services are also considered in lieu of cash contributions. This multi-year industry commitment not only stabilizes the finances of the AMRC but also provides a funding annuity that AMRC can utilize as industry matching funding to pursue additional UK and European Union research grants.

The AMRC utilizes two boards, both led by a chair elected by the membership. The AMRC industrial board is focused on the overall management and strategic direction of the AMRC and is populated by a single representative from each Tier 1 member and fractional representation of Tier 2 members. This same tier-based board seat approach is utilized in the AMRC technical board, where decisions regarding the research agenda for the collaborative are made and specific research projects are proposed, voted upon and approved depending upon the support of the industry members. Research funding is drawn from the annual contributions of the industry members and research is undertaken at a precompetitive level, solving problems common to many of the industry members.

One success factor is well-defined rules for ownership and use of IP in collaboration. The precompetitive research performed by the AMRC creates foreground IP that is owned by the University of Sheffield and licensed to the industry members, royalty free, for their use and commercial exploitation in their business and within their supply chain. This approach is typically well received by industry members, since the IP generated by their annual contributions is held by a noncommercial, neutral third-party (the university) and no individual industry member holds more control over this common-benefit IP than another.

In order to move from precompetitive research to product specific research, the collaboration model includes the ability for industry members to sponsor additional projects that address specific product needs. These projects build upon the body of precompetitive IP (which all members have rights to utilize) but involve the sponsoring industry member's proprietary information. As such, these project agreements are established and financed separately from the collaboration agreement and include foreground IP ownership for the sponsoring industry member, since this IP represents a competitive advantage in the market. For example, one special project dramatically increased the machining efficiency of a family of turbine engine blade mounting hubs for a new product line.

Ten years into the growth of the AMRC, major industry partners started to establish manufacturing plants on the AMRC campus. In anticipation of a need to expand the skilled workforce for both new and existing business, an apprentice training center was established on the AMRC campus. The training center admits 200–250 apprentices each year and typically receives 1000 applications each year for the available positions. Each apprentice is sponsored by a local manufacturer, who funds half of their tuition, with the other half provided by the UK government. The apprentices undertake a three-year program, with their time split between training at the AMRC and working at their sponsoring company. The apprentices receive a modest salary while in the program and graduate debt free, with a full-time job waiting for them. Two tracks are available, one being strictly a path to apprentice certification and entry to a full-time position at their sponsoring company. The second track includes additional academic training, so the apprentice has the opportunity to continue their education at the University of Sheffield securing the equivalent of a two- or four-year degree. If so inclined, the individual can work all the way to a MS or PhD degree. Having an apprentice training center to PhD degree program housed

at a top research university is unique to the University of Sheffield and has proven highly effective.

6.5.1 *Lessons Learned*

The AMRC model has been replicated numerous times globally. Each new collaborative had a unique technical focus, but the model described above was generally followed. A series of lessons learned emerged from this work, which are summarized below.

Find the Common “Pain” in Industry Perhaps the most important decision in conceptualization of a research collaborative is selection of the technical theme. By focusing on common pain in industry, the technical theme shapes itself. For example, at the start of the AMRC, the technical theme was simply the machining of hard metals. Many companies were struggling with the machining of hard metals and independently making significant research investments to try to improve their processes. A good way to start identifying a theme for the consortium is to bring the proposed launch members together to discuss their biggest problems. The key is to find a theme for the consortium that has a clear and simple value proposition for each launch member. Ask the question, what are the launch members spending money on independently that could be addressed with a shared investment? The answer to this question is the “pain” in industry and a potential research theme for the consortium.

Maintain a Narrow Technology Focus to Start Once identified, the theme will need to be narrowed, so the investment of the initial launch members and government co-investment will be impactful enough to deliver near-term results. If the technology focus is too broad, the consortium will spread its investment so thin across a range of problems that it will be unable to deliver results on any one project. Returning to the AMRC, the initial focus was “machining of hard metals.” If the focus had just been machining broadly, one can imagine the wide range of projects that would have emerged. By narrowing the focus specifically to *hard metals* machining, the problem set was limited and the selection of initial projects was a manageable task. With the initial financial resources focused on a few problems, the AMRC was able to set clear success metrics and quickly deliver results and benefits to the consortium members. This success drew in new members, new government grants and ultimately allowed the AMRC to expand beyond their initial narrow focus to additional technical areas.

While a narrow focus is advisable to start a consortium, choosing too narrow a technical focus can limit your ability to grow membership over time, since the benefits may be restricted to a small industry segment. For example, suppose the initial focus of the AMRC had been development of electrostatic-discharge machining techniques. In that case, turbine engine builders may have been interested, since

electrostatic-discharge machining is important to their production. On the other hand, aircraft builders may not have been interested in joining the consortium. Identifying the technical focus requires a balanced approach to ensure it is narrow enough to allow effective investment but not so narrow as to limit the ability to secure commitment from the launch members and grow the consortium over time.

Limit the Number of Launch Members This recommendation seems counter intuitive. Why would you want to turn away a company that is ready to commit and pay annual fees and be part of the collaborative? Experience has shown that when the number of launch members grows larger than four to five companies plus the host organization, gaining consensus on the collaboration agreement among the launch partners can be difficult. If the number of launch members grows too large, the position of each member in the supplier value chain should be considered. Are they filling a gap in the value chain framework? In addition, is it a well-known company, whose brand will draw in other members in the future? These are two important questions to ask, if too many companies are interested in being launch members. It is important to communicate to the companies not selected for the launch that they will have just as important a role in the consortium once it is up and running. The only privilege they will not enjoy is the opportunity to participate in shaping the collaboration agreement. However, non-launch members can be assured that the launch members will craft an agreement that is balanced and does not put the needs of any single member above the others. With a small number of launch members, the collaboration agreement can be more efficiently negotiated. Once signed, the consortium can immediately add new members, assuming the resulting collaboration agreement is to their liking.

Consortium Members Should Span the Supplier Value Chain One reason this collaborative model can deliver results is the ability to recruit members that span the supplier value chain of the technical area being researched. When the full value chain jointly decides what problems to address and innovation occurs, it is a short path to market since all the players required to transition the technology to production were involved from the start. Perhaps the best example of a full value chain at consortium launch is the Thermoplastic Composite Research Center (TPRC) in The Netherlands. The launch members were Boeing (an Original Equipment Manufacturer), Fokker (a major aero-structures fabricator [now GKN Fokker] and supplier to Boeing) and TenCate (a thermoplastic composite material supplier). With an OEM, aero-structures supplier and material supplier as the launch members, the framework of the full thermoplastic composites value chain was participating. With the launch of the consortium, the next task was to fill out this framework by bringing in additional companies, such as equipment builders, inspection systems companies and assembly automation specialists.

Create an Environment to Allow Competitors to Work Together It may seem like an impossible task to get competitors to work together in a consortium setting, but experience shows that it is possible if the right conditions are established. It is

important that precompetitive research is performed with financial resources collected as annual membership fees and government grants. The university owns the IP represented by this precompetitive research and jointly utilizes it through licensing by all members without any one member feeling that they are giving up a competitive advantage. This precompetitive research has limited value to industry, unless there is a path for it to underpin product specific research that transitions the innovation taking place in the consortium to the members' products or services. For this to occur, the consortium agreement should include the ability to perform additional projects that transition a particular precompetitive innovation to a product or service specific application. In contrast to joint access by members to the precompetitive IP, when a member sponsors an additional project, they negotiate ownership rights of the resulting IP. These additional projects are the technology transition vehicle for the consortium. Members that sponsor more product specific projects often reap the most benefit from the consortium.

The Host Organization The host organization is critical to the success of the consortium. The host represents neutral ground, where industry members can gather to perform collaborative research and typically performs most of the research sponsored by the consortium. In most cases, the host is a university with a strong research reputation in the selected technical focus of the research consortium plus it has a recognized history of working well with industry. The host also plays an important role as owner of the precompetitive IP created in the consortium. This ownership of IP by the host brings two benefits. First, no one industry member holds the IP; however, all members have access to the IP via their membership in the consortium. Secondly, the creation of new IP that resides at the host organization is a strong incentive for investment by the government of the host organization.

Secure Multi-Year Commitment with Co-Investment by Government With the complexity of launching a consortium, including building up staff and getting research projects underway, progress in the early stages can be slow. As such, it is important that the launch members commit to a multi-year engagement with the consortium to ensure a solid financial footing in the challenging start-up period. Typically, a minimum commitment period of three to five years is recommended, with an option for leaving the consortium under extreme circumstances. A key factor in securing a multi-year commitment from the launch members is to secure government co-investment at the national, state and/or regional level. This government co-investment combined with co-investment by other launch members can produce a strong business case to help companies secure approval from their management to join the consortium.

Select the Right Leadership at the Host Organization Important leadership attributes include creativity, team building and the ability to share a concise and compelling value proposition to grow membership. Additionally, the leadership should have extensive experience with university and government technology investment policy and processes to secure future grants and grow the consortium. In

the early stages of the consortium, a single individual from the host organization often performs both technical and commercial director roles to reduce start-up costs. Often the person identified for this role is an existing senior professor or researcher from the host organization. It is important that the person selected can effectively operate at high levels in academic, industry and government settings. This assignment will be a full-time job. Launching and growing a consortium cannot be successfully accomplished on a part-time basis. Finally, to strengthen the consortium leadership over time, there should be both full-time technical and commercial directors, with the technical director focused on ensuring high quality, impactful research results for industry members and the commercial director focused on strengthening relationships with existing members and recruiting new members.

Make Infrastructure Investments In-Line with Government Objectives With a wide range of collaborative industry research centers launched around the world, there has been an equally diverse range of approaches to creating the required infrastructure and facilities. At one extreme is the AMRC at the University of Sheffield. In 2001, it was important for the UK government to make a statement about renewal of industry in Northern England, deciding to invest millions of GBPs in green field infrastructure. Shiny new buildings located at the site of a shuttered open coal mine allowed the UK government to show that it was investing for the future. The SERC Aerospace Program in Singapore is at the other extreme. With this consortium hosted by the Singapore government's A*STAR lab, existing research facilities were initially used to perform the research and to serve as a gathering point for industry members. They had already invested in the infrastructure and simply repurposed it to support the objectives of the consortium. This was a more affordable approach to launching a consortium. Considering these examples, the formative stage should address two key questions: (1) what are the objectives of the government in establishing a research consortium? and (2) where in this range of infrastructure investment approaches can these objectives be most readily met?

Leverage the Brand Value of Members Once the launch members are selected and an agreement is signed, growing membership and filling out the supplier value chain is important to the sustainability and success of the consortium. The University of Sheffield had this in mind when they asked Boeing to co-brand the AMRC, thus the name The Advanced Manufacturing Research Center with Boeing. The AMRC used the Boeing logo on their business cards, signage, letterhead and team wear. When the AMRC was launched, Boeing was the first and only member. By using the power of the Boeing brand, the AMRC was able to reach out to a wide range of potential members and paint a picture of how they could strengthen their relationship with Boeing while they secured affordable, world class research services. The brand power of Boeing and other OEMs, like Rolls Royce and BAE Systems, have been important to the growth and success of the AMRC.

6.6 Takeaways and Future Strategies

Companies and universities can collaborate in a myriad of ways. In most strategic UI partnerships, the company and university have shared goals and jointly pursue opportunities. In UI consortia, the higher the investment and commitment from the partners, the more that the value proposition needs to speak to the needs and interests of the participating organizations. Active management of strategic or multifaceted collaborative relationships is critical to the success and longevity of the relationship and to ensuring mutual benefits to all parties. Likewise, the governance structure, supporting legal agreements, and IP approach should all align with the goals of the consortium, the level of investment, the stage of research (basic vs. applied) and the research domain.

Though we have focused in this chapter on alliances and consortia initiated by universities, because they are the most common, the principles presented also apply to partnerships initiated by individual companies, groups of companies, nonprofit organizations or governmental entities. A research center, for example, might be an entity within a university, as discussed in this chapter, be governed by an external entity, or possibly by a venture jointly held by the center's participants.

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